23 (a) Show that the equation $2x^3 + 4x = 3$ has a solution between 0 and 1

(2)

(b) Show that $2x^3 + 4x = 3$ can be rearranged to give $x = \frac{3}{4} - \frac{x^3}{2}$

(1)

(c) Starting with $x_0 = 0$, use the iteration formula $x_{n+1} = \frac{3}{4} - \frac{x_n{}^3}{2}$ three times to find an estimate for the solution to $2x^3 + 4x = 3$

.....

(3)

(Total for Question 23 is 6 marks)